$$\sum_j \dot{\alpha}_j(t)|E_j\rangle = -\frac{i}{\hbar}\mathbf{H}\sum_j \alpha_j(t)|E_j\rangle.$$

Next, we use the fact that $\mathbf{H}|E_j\rangle = E_j\,|E_j\rangle$ to get

$$\sum_j \dot{\alpha}_j(t)|E_j\rangle = -\frac{i}{\hbar}\sum_j E_j\alpha_j(t)|E_j\rangle$$

or, regrouping,

$$\sum_j \left\{\dot{\alpha}_j(t) + \frac{i}{\hbar}E_j\alpha_j(t)\right\}|E_j\rangle = 0.$$

The final step should be easy to see. If a sum of basis vectors equals zero, every coefficient must be zero. Hence, for each eigenvalue E_j, $\alpha_j(t)$ must satisfy the simple differential equation

$$\frac{d\alpha_j(t)}{dt} = -\frac{i}{\hbar}E_j\alpha_j(t).$$

This, of course, is the familiar differential equation for an exponential function of time, in this case with an imaginary exponent. The solution is

$$\alpha_j(t) = \alpha_j(0)e^{-\frac{i}{\hbar}E_j t}. \qquad (4.30)$$

This equation tells us how the α_j change with time. It is quite general and not restricted to spins, provided that the Hamiltonian does not depend explicitly on time. This is our first example of the deep connection between energy and frequency, which recurs over and over throughout quantum mechanics and quantum field theory. We will return to it often.